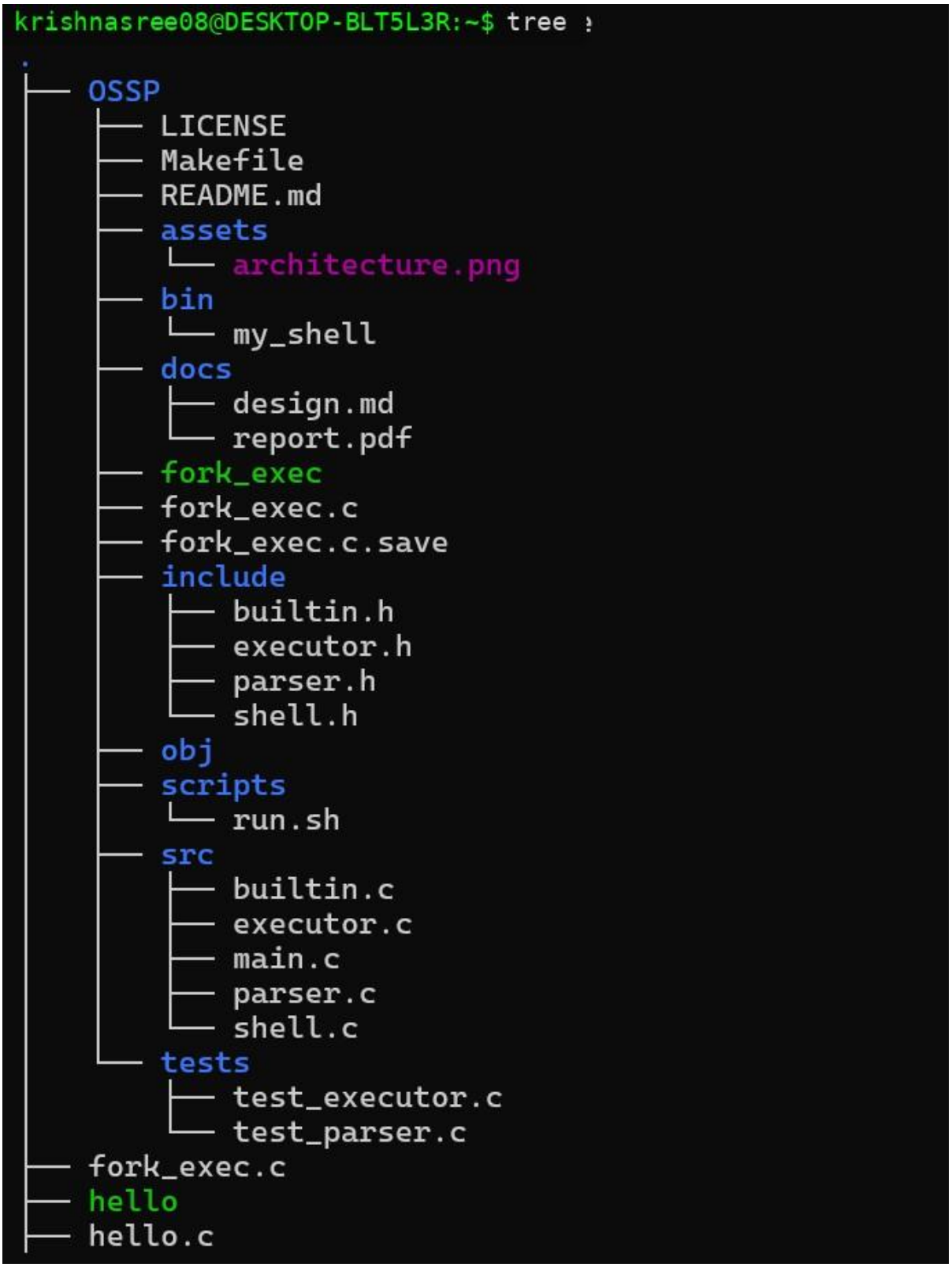
SKILL (Week 2)

Session -1

Task1: To Install Linux VM, Configure GCC, Setup Git Repository, Create Project Structure, Understand Shell Architecture, Build Initial Makefile.



Observation: I explored the Linux development environment and practiced organizing a shell project into appropriate directories and files. I also learned the role of GCC in compiling C programs and understood how Git helps in maintaining and tracking the project files. This activity helped me develop a clear understanding of how a software project is structured and managed systematically.

Task 2:

To Analyse process abstraction, execute fork(), understand exec() family, analyse parent-child relationships, inspect process tree, and practice system call tracing.

```c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    char command[100];

    printf("Enter a Linux command: ");
    scanf("%s", command);

    pid_t pid = fork();

    if(pid < 0)
    {
        printf("Fork failed!\n");
    }
    else if(pid == 0)
    {
        printf("Child Process\n");
        printf("Child PID: %d\n", getpid());

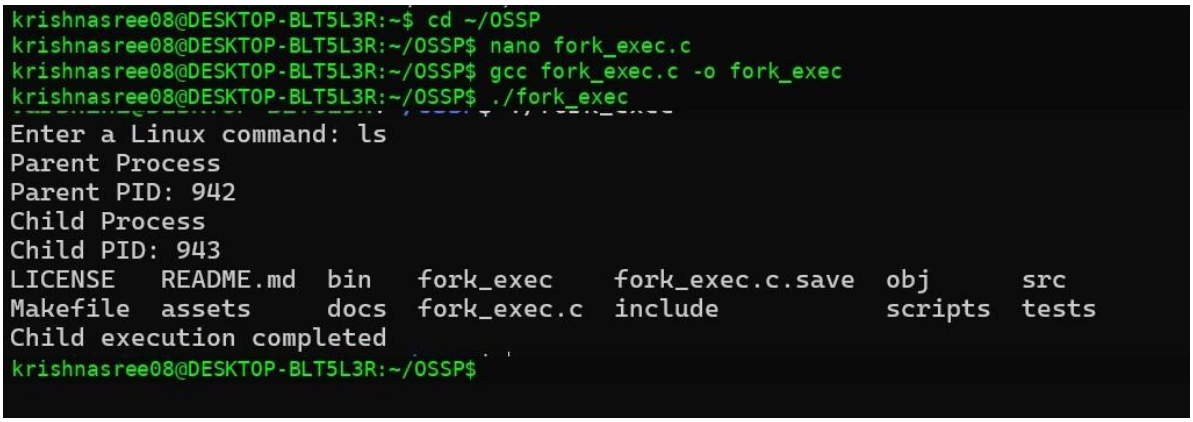
        execlp(command, command, NULL);

        printf("Execution failed\n");
    }
    else
    {
        printf("Parent Process\n");
        printf("Parent PID: %d\n", getpid());

        wait(NULL);

        printf("Child execution completed\n");
    }

    return 0;
}
```


krishnasree08@DESKTOP-BLT5L3R:~$ cd ~/OSSP
krishnasree08@DESKTOP-BLT5L3R:~/OSSP$ nano fork_exec.c
krishnasree08@DESKTOP-BLT5L3R:~/OSSP$ gcc fork_exec.c -o fork_exec
krishnasree08@DESKTOP-BLT5L3R:~/OSSP$ ./fork_exec
Enter a Linux command: ls
Parent Process
Parent PID: 942
Child Process
Child PID: 943
LICENSE README.md bin fork_exec fork_exec.c.save obj src
Makefile assets docs fork_exec.c include scripts tests
Child execution completed
krishnasree08@DESKTOP-BLT5L3R:~/OSSP$